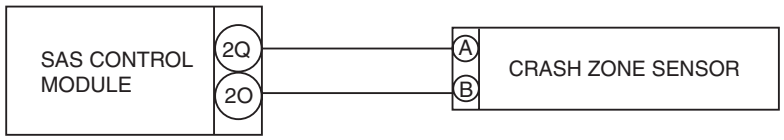
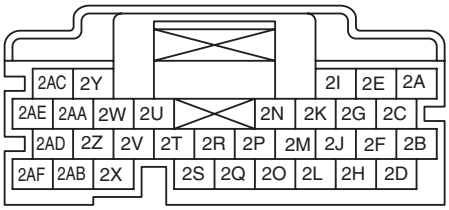
DTC B0094:11/B0094:12/B0094:29/B0094:2B/B0094:4A/B0094:87/B0094:96 [TWO-STEP DEPLOYMENT CONTROL SYSTEM]

id0802b2813400

System malfunction location	• B0094:11: Crash zone sensor circuit short to body ground • B0094:12: Crash zone sensor circuit short to power supply • B0094:29, B0094:96: Crash zone sensor internal malfunction • B0094:2B: Short circuit to crash zone sensor and other air bag sensor circuits • B0094:4A: Crash zone sensor ID error (assembly incorrect) • B0094:87: Signal reception error from crash zone sensor
Detection condition	**Warning** **• Detection conditions are for understanding the DTC outline before performing an inspection. Performing an inspection according to only the detection conditions may cause injury due to an operating error, or damage the system. When performing an inspection, always follow the inspection procedure.** • Wiring harness between the crash zone sensor and SAS control module has a malfunction • Crash zone sensor has a malfunction
Fail-safe function	Not applicable
Possible cause	• Crash zone sensor assembly incorrect • Crash zone sensor connector malfunction • Short circuit in the terminal or wiring harness between the crash zone sensor and SAS control module • Short circuit to power supply or body ground in the wiring harness between the crash zone sensor and SAS control module • Open circuit in the wiring harness between the crash zone sensor and SAS control module • Crash zone sensor malfunction • SAS control module malfunction

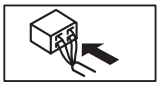


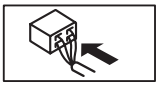
SAS CONTROL MODULE WIRING HARNESS-SIDE CONNECTOR



CRASH ZONE SENSOR WIRING HARNESS-SIDE CONNECTOR







Diagnostic Procedure

Step	Inspection		Action
1	**INSPECT CRASH ZONE SENSOR** **Warning** **• Handling the component parts improperly can accidentally operate (deploy) the air bag modules and pre-tensioner seat belts, which may seriously injure you. Read the service warnings/cautions and the workshop manual before handling the air bag system components. (See AIR BAG SYSTEM SERVICE WARNINGS [STANDARD DEPLOYMENT CONTROL SYSTEM].) (See AIR BAG SYSTEM SERVICE CAUTIONS [STANDARD DEPLOYMENT CONTROL SYSTEM].)** • Switch the ignition off. • Disconnect the negative battery cable and wait for **1 min or more**. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Verify that the crash zone sensor is correctly installed. (See CRASH ZONE SENSOR REMOVAL/ INSTALLATION [STANDARD DEPLOYMENT CONTROL SYSTEM].) • Is the crash zone sensor correctly installed?	Yes	Go to the next step.
		No	Correctly install the crash zone sensor. (See CRASH ZONE SENSOR REMOVAL/ INSTALLATION [STANDARD DEPLOYMENT CONTROL SYSTEM].) Then go to the next step.
2	**INSPECT CRASH ZONE SENSOR CONNECTOR** • Switch the ignition off. • Disconnect the negative battery cable and wait for **1 min or more**. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Remove the front bumper. (See FRONT BUMPER REMOVAL.) • Disconnect the crash zone sensor connector. (See CRASH ZONE SENSOR REMOVAL/ INSTALLATION [STANDARD DEPLOYMENT CONTROL SYSTEM].) • Inspect the crash zone sensor connector. (Corrosion, damage, and disconnected pins) • Is there any malfunction of the crash zone sensor connector?	Yes	Replace the malfunction part, then go to step 6.
		No	Go to the next step.

Step	Inspection		Action
3	**INSPECT CRASH ZONE SENSOR CIRCUIT FOR SHORT TO GROUND** • Remove the column cover. (See COLUMN COVER REMOVAL/ INSTALLATION.) • Disconnect the clock spring connector. (See CLOCK SPRING REMOVAL/ INSTALLATION [STANDARD DEPLOYMENT CONTROL SYSTEM].) • Remove the glove compartment. (See GLOVE COMPARTMENT REMOVAL/ INSTALLATION.) • Disconnect the passenger-side air bag module connector. (See PASSENGER-SIDE AIR BAG MODULE REMOVAL/INSTALLATION [TWO-STEP DEPLOYMENT CONTROL SYSTEM].) • Disconnect the driver and passenger-side front seat connectors. (See FRONT SEAT REMOVAL/ INSTALLATION.) • Remove the B-pillar lower trim. (See B-PILLAR LOWER TRIM REMOVAL/ INSTALLATION.) • Disconnect the driver and passenger-side pre-tensioner seat belt connectors. (See FRONT SEAT BELT REMOVAL/ INSTALLATION.) • Remove the headliner. (See HEADLINER REMOVAL/ INSTALLATION.) • Disconnect the driver and passenger-side curtain air bag module connectors. (See CURTAIN AIR BAG MODULE REMOVAL/ INSTALLATION [STANDARD DEPLOYMENT CONTROL SYSTEM].) • Disconnect the all SAS control module connectors. (See SAS CONTROL MODULE REMOVAL/ INSTALLATION [STANDARD DEPLOYMENT CONTROL SYSTEM].) • Inspect for continuity between the following terminals (wiring harness-side) and body ground: — SAS control module terminal 2Q — SAS control module terminal 2O **Note** • Inspect for continuity while shaking the wiring harness between the SAS control module and crash zone sensor. • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between SAS control module terminal and crash zone sensor terminal. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Replace the malfunctioning part. **If there is no common connector:** • Replace the wiring harness which has a short to ground. Go to Step 6.
		No	Go to the next step.

Step	Inspection		Action
4	**INSPECT CRASH ZONE SENSOR CIRCUIT FOR OPEN CIRCUIT** • Crash zone sensor and SAS control module connectors are disconnected. • Inspect for continuity between the following terminals (wiring harness-side): — Crash zone sensor terminal A—SAS control module terminal 2Q — Crash zone sensor terminal B—SAS control module terminal 2O **Note** • Inspect for continuity while shaking the wiring harness between the SAS control module and crash zone sensor. • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between SAS control module terminal and crash zone sensor terminal. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Replace the malfunctioning part. **If there is no common connector:** • Replace the wiring harness which has an open circuit. Go to Step 6.
5	**INSPECT CRASH ZONE SENSOR CIRCUIT FOR SHORT TO POWER SUPPLY** • Crash zone sensor and SAS control module connectors are disconnected. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off or on). • Measure the voltage at the following terminals (wiring harness-side): — SAS control module terminal 2Q — SAS control module terminal 2O **Note** • Measure the voltage while shaking the wiring harness between the SAS control module and crash zone sensor. • Is the voltage **0 V**?	Yes	Replace the crash zone sensor, then go to the next step. (See CRASH ZONE SENSOR REMOVAL/ INSTALLATION [STANDARD DEPLOYMENT CONTROL SYSTEM].)
		No	Refer to the wiring diagram and verify whether or not there is a common connector between SAS control module terminal and crash zone sensor terminal. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Replace the malfunctioning part. **If there is no common connector:** • Replace the wiring harness which has a short to power supply. Go to the next step.
6	**PERFORM SAS CONTROL MODULE DTC INSPECTION** • Switch the ignition off. • Disconnect the negative battery cable and wait for **1 min or more**. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Connect the SAS control module connectors. • Reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off or on). • Clear the DTC for the SAS control module using the M-MDS. (See CLEARING DTC [STANDARD DEPLOYMENT CONTROL SYSTEM].) • Perform the DTC inspection for the SAS control module using the M-MDS. (See DTC INSPECTION [STANDARD DEPLOYMENT CONTROL SYSTEM].) • Are the same DTCs present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the SAS control module. (See SAS CONTROL MODULE REMOVAL/ INSTALLATION [STANDARD DEPLOYMENT CONTROL SYSTEM].)
		No	DTC troubleshooting completed.